CS 435: Project Report Instructions

Each team submits one project report, which is worth 15% of your overall grade. This document specifies what information you should include in your report. It requires following the IEEE template, which is available on D2L.

1 Report contents (6-8 pages)

Your final report should be written in the same style as an NLP research paper**.** Your report should be 6-8 pages (excluding references). Consider using the following section structure, though you can use a different structure.

Key information. Your report should have the following information:

- **Title**: The title of your project.
- **Team member names**: List the names of all of your team members.

Abstract. An abstract should concisely (less than 300 words) motivate the problem, describe your aims, describe your contribution, and highlight your main findings.

Introduction. The introduction explains the problem, why it's difficult, interesting, or important, how and why current methods succeed/fail at the problem, and explains the key ideas of your approach and results. Though an introduction covers similar material as an abstract, the introduction gives more space for motivation, detail, references to existing work, and to capture the reader's interest.

Related work. This section helps the reader understand the research context of your work, by providing an overview of existing work in the area.

- You might discuss: papers that inspired your approach, papers that you use as baselines, papers proposing alternative approaches to the problem, papers applying your methods to different tasks, etc.
- This section shouldn't go into deep detail in any one paper (for example, there probably shouldn't be any equations) – instead it should explain how the papers relate to each other, and how they relate to your work.
- Attempt to demonstrate, as you review the literature, limitations or motivations that point to why *your* work is a nice next step, or useful replication, or promising analysis (or otherwise, if your work doesn't fall into these categories!).

Approach. This section details your approach(es) to the problem. For example, this is where you describe the architecture of your neural network(s), and any other key methods or algorithms.

- You should be specific when describing your main approaches – you prob-

ably want to include equations and figures.

- You should also describe your baseline(s). Depending on space constraints and how standard your baseline is, you might do this in detail, or simply refer the reader to some other paper for the details. Default project teams can do the latter when describing the provided baseline model.
- If any part of your approach is original, make it clear (so we can give you credit!). For models and techniques that aren't yours, provide references.
- If you're using any code that you didn't write yourself, make it clear and provide a reference or link. When describing something you coded yourself, make it clear (so we can give you credit!).

Experiments. This section contains the following.

- **Data**: Describe the dataset(s) you are using (provide references). If it's not already clear, make sure the associated task is clearly described. Being precise about the exact form of the input and output can be very useful for readers attempting to understand your work, especially if you've defined your own task.
- **Evaluation method**: Describe the evaluation metric(s) you use, plus any other details necessary to understand your evaluation. Some projects will have clear metrics from prior work on given datasets, but we realize that other projects will define their own metrics. If you're defining your own metrics, be clear as to what you're hoping to measure with each evaluation method (whether quantitative or qualitative, automatic or human-defined!), and how it's defined.
- **Experimental details**: Report how you ran your experiments (e.g. model configurations, learning rate, training time, etc.)
- **Results**: Report the quantitative results that you have found so far. Use a table or plot to compare results and compare against baselines. [3]
  - If you're a default project team, you should **report the F1 and EM scores you obtained on the test leaderboard** in this section. Make it clear whether you are on the IID SQuAD track or the RobustQA track. You can also report dev set results if you like.
  - Comment on your quantitative results. Are they what you expected? Better than you expected? Worse than you expected? Why do you think that is? What does that tell you about your approach?

Analysis. Your report should include *qualitative evaluation*. That is, try to understand your system (e.g. how it works, when it succeeds and when it fails) by inspecting key characteristics or outputs of your model.[4]

[2]In experimental work, this could mean giving a specific name to each method, each dataset, each baseline; it could also mean making consistent use of mathematical notation where appropriate.

[3]Some analysis-centric custom projects will not have model-based baselines to compare against; in this case, consider whether there are simpler analysis methods than your proposed method which might achieve the same goal. It would be great to evaluate them, or at least

discuss why they are insufficient compared to your method. Even in analysis projects, it's important to ask whether we could be achieving similar insights using simpler ("baseline") methods. So, if you don't have a baseline analysis method to compare against, discuss some alternative methods of analysis that one could use, and why (mathematically, linguistically, or otherwise) you believe your proposed method is superior.

[4]For some analysis-centric custom projects, it might seem odd to have a separate analysis section. It's up to you whether this section is included explicitly, but it might be a good idea.

Conclusion. Summarize the main findings of your project and what you have learnt. Highlight your achievements, and note the primary limitations of your work. If you like, you can describe avenues for future work.

References. Your references section should be produced using BibTeX.

Team contributions. If you are a multi-person team, add a brief summary of what each team member did for the project (about 1 or 2 sentences per person). We will read these descriptions. For almost all teams, it will have no effect (i.e., team members all receive the same grade), but for teams with considerably unequal contributions, we may investigate and/or give different grades to team members.

2 Improving your technical writing

As a reminder, the *Milestone Instructions* included extensive resources to help you improve your technical writing. You can use these, and previous feedback you've received, to improve your technical writing.

3 Grading and feedback

Your project report will be graded holistically, taking into account many criteria: originality, performance of your methods, complexity of the techniques you used, thoroughness of your evaluation, amount of work put into the project, analysis quality, writeup quality, demonstrating strong understanding, etc.

4 Code

We ask you to submit your code as a zip file (up to 1MB) to D2L.

Your code will not be graded.